[TODO: title] — Near Real-Time Markerless Scoring of the Standardized Drinking Task

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Abstract—[TODO: Write last, ~200 words. Structure: (1) OMC plus biomechanical modeling is the reference standard for upper-limb movement quality but is lab-bound and slow; (2) markerless pipelines have closed much of the accuracy gap, but the differentiable-biomechanics approach needs days of GPU optimization per cohort, so results never reach the clinician during the visit; (3) we present a markerless pipeline — multi-camera 2D pose, robust bundle adjustment, temporal smoothing, geometric measure extraction — that scores the standardized drinking task within X of the end of recording on a single consumer GPU; (4) validated on 11 participants and 697 trials against synchronized optical mocap, we reach $r_s \geq 0.86$ on six of eleven Murphy measures and $r \geq 0.96$ on elbow-extension trajectories, without inverse kinematics; (5) accuracy is comparable to offline markerless work while running X× faster, making same-session feedback feasible.]

Index Terms—markerless motion capture, optical motion capture, drinking task, movement quality, stroke rehabilitation, real-time

I. Introduction

[TODO: P1 — clinical need: stroke prevalence, upper-limb impairment, the case for frequent objective assessment of movement quality.]

[TODO: P2 — the standardized drinking task and the Murphy measures as the established instrument [1]; OMC plus biomechanical modeling as the reference standard.]

[TODO: P3 — markerless MMC now approaches OMC accuracy [2], removing markers and specialist labor. State the remaining obstacle plainly: latency. Give the concrete figure — the end-to-end differentiable pipeline of [2] required roughly 12 days of GPU processing for 1160 trials, dominated by 2D keypoint extraction over 5800 videos, plus 30 CPU hours of OpenSim IK on the OMC side. A result that arrives days later cannot inform the session that produced it.]

[TODO: P4 — our position. Same task, same measures, same validation protocol, but every stage chosen so per-trial compute is bounded and overlaps with recording. Note the deliberate trade: we compute measures geometrically from 3D keypoints rather than fitting a biomechanical model, which costs some trajectory agreement and buys the latency.]

Contributions:

- A markerless pipeline that produces the Murphy drinking-task movement-quality measures in near real time on a single consumer GPU, with [TODO: a measured per-stage latency budget].
- A validation against synchronized optical mocap on 11 participants and 697 trials, reporting per-trajectory and per-measure agreement (Tables I–III).
- [TODO: An account of which measures survive the no-IK, geometric construction and which do not — and why the failures are anatomical (occluded seated hips) rather than algorithmic.]

II. Related Work

A. Movement quality assessment in the drinking task

[TODO: Murphy et al. measures, clinical use, what each is meant to capture.]

B. Markerless motion capture for clinical kinematics

[TODO: Multi-view pose estimation, triangulation, biomechanical fitting. Position [2] as the closest comparison: same task, same measures, OMC-validated — but offline, and IK-based.]

C. Latency in clinical measurement pipelines

[TODO: Little prior work reports latency at all. Argue that time-to-result is a first-class metric for clinical adoption, and that reporting accuracy alone has hidden this axis.]

III. Methods

A. Participants and Task

[TODO: DELTA cohort: 11 participants, both arms, standardized drinking task. Fill in demographics, impairment scores, ethics approval, repetitions per arm. State the trial accounting — 697 trials enter the measure comparison; the per-frame trajectory comparison (Tables I, II) is computed on the 220-trial subset for which per-frame OMC trajectories were exported — and say why, or extend the export.]

B. Data Acquisition

[TODO: Camera rig: number of cameras, model, resolution, frame rate (60 Hz), geometry, ChArUco calibration, synchronization. OMC system and marker set, sampling rate (100 Hz). Note the camera-quality auditing that had to precede any of this — mis-cut clips and moved cameras, and how they were detected — since it bears directly on what a deployable system must self-check.]

C. Pipeline

[TODO: One paragraph per stage, each stated with its throughput, because cost is the point of the paper:]

1) 2D body pose. An off-the-shelf YOLO-pose network, run per frame on every camera. [TODO: state version, input size 640 — and why inference must run at the size the network was trained at — and the fps achieved.]
2) Cup localization. A separate YOLO detector locates the cup once, at the start of the trial; a transformer visual-object tracker (UETrack-B) then follows it through the remaining frames in each camera. The cup trajectory is used only to segment the movement into phases (Sec. III-D); no movement-quality measure is computed from it. [TODO: Motivate detect-once: per-frame detection loses the cup at the drinking apex, where it is occluded and small — several 1–3 s blackouts per trial, falling exactly on the phase that defines the segmentation. Detect-once plus tracking fills those frames. State that no re-anchoring is used, and why: re-seeding from a shared consensus couples the cameras and can cascade them all onto the same wrong object, destroying the independence that multi-view voting relies on.]
3) 3D reconstruction. Robust-reprojection bundle adjustment over all nine upper-body joints, Huber loss, no bone-length prior. [TODO: Say explicitly that a bone-length prior was tested and rejected: it rigidifies bone lengths but slides joints along the unobservable depth ray, worsening agreement with OMC. Describe the multi-view consensus rule used to accept a 3D point.]
4) Temporal smoothing. SmoothNet. [TODO: State that smoothing, not the reconstruction, is what governs peak-velocity accuracy.]
5) Phase segmentation. From the cup trajectory; see Sec. III-D.
6) Measure extraction. Geometric, from 3D body keypoints only; see Sec. III-E.

D. Phase Segmentation

[TODO: Describe the segmenter: how the cup 3D trajectory yields the drinking-task phases (speed and displacement-from-rest), and the dwell definition. This is the only place the cup is used.]

[TODO: Keep the two roles of the cup distinct and say so plainly: the segmentation reported in the deployed pipeline comes from the cup, but the measure validation in Sec. IV deliberately uses the OMC phases for both systems, so that the numbers isolate pose error from phase-classification error. If segmentation accuracy is to be claimed, it needs its own comparison against OMC phase boundaries — reported as boundary error in seconds, separately from the measure tables.]

[TODO: Note on the cup's absolute accuracy, if any cup number is reported: our centroid is the cup body while the mocap markers sit on the rim, giving a constant ≈ 60 mm offset, and there is a displacement-proportional $\approx 10\%$ scale error that belongs to the camera calibration, not the tracker. The latter is not cup-specific — it inflates every 3D distance and velocity the pipeline reports, body keypoints included — so decide whether it is disclosed here or in Limitations.]

E. Movement Quality Measures

[TODO: Condense measures_methods.md into publishable prose. The two ground rules to state up front: (i) smooth position, then differentiate — never filter a raw derivative; (ii) all peak/angle measures are windowed by phase. Then the constants table (4 Hz position low-pass, 6 Hz on the speed for the peak-velocity operator, 2nd-order zero-phase Butterworth, 60 mm/s movement-unit amplitude, 0.15 s gap) and the geometric definition of each angle. Justify the three choices that are non-obvious and were empirically forced: peak velocity uses the differentiate-then-low-pass operator and is scoped to reaching; trunk displacement is a 3D excursion of the shoulder midpoint rather than an anatomical forward projection, because the forward normal depends on occluded seated hips; shoulder flexion uses a per-trial constant trunk-down axis for the same reason.]

[TODO: Also state which measures are not reported and why: the two percent-timing variants (redundant once phase duration is shared), and drinking-phase shoulder flexion (no ground-truth column exists), giving 11 panels rather than 12.]

F. Latency Model

[TODO: THIS SECTION IS THE PAPER'S NOVELTY AND IS NOT YET MEASURED. Define precisely what is timed — wall-clock from the last recorded frame of a trial to its scored output — and decompose it per stage. Separate streaming cost, which overlaps with recording, from tail cost, which can only start once the trial ends; the headline claim is a bound on the tail. Specify the hardware and state whether recording and inference share the GPU. Make the definition reproducible enough that another group can time their own pipeline the same way.]

G. Validation Protocol

All measures are computed from our markerless keypoints alone; the optical system's processed output is read only as the value to compare against and never enters our computation. [TODO: Spatial and temporal alignment to OMC; time lag from wrist-speed cross-correlation; static-bias handling. Per-trajectory agreement as r, RMSE, bias and lag; per-measure agreement as Spearman r_s across trials and r_{av} on per-participant, per-arm averages. State that correlations are computed within-arm and never pooled across affected and unaffected arms, since pooling would inflate r with the between-arm difference.]

[TODO: Important framing point to make here, not buried in Discussion: for a measure whose OMC values are near-constant across a cohort, a rank correlation has little true variance to track and reads low even when the two systems agree. Report such measures as agreement (fraction of trials within a tolerance, plus bias), and check the OMC-side effect size before drawing any conclusion from a low r.]

IV. Results

A. Study Population

[TODO: Table: demographics, impairment scores, trial counts, exclusions with reasons. 826/837 recorded repetitions were scorable.]

B. Latency

[TODO: Table: per-stage wall-clock, streaming versus tail, total time-to-result, and the hardware. Compare against the offline baseline normalized to the same trial count. NOT YET MEASURED — this table does not exist and is the main gap between the current results and the paper's claim.]

C. Kinematic Trajectory Agreement

Table I reports per-frame agreement for the six kinematic trajectories, and Table II the within-participant variability of the static bias. Fig. 1 shows exemplary trajectories for one participant. [TODO: Prose: elbow extension and shoulder flexion agree closely; the velocity trajectories sit near $r = 0.75$; shoulder abduction on the unaffected arm is the weakest. Say what the bias columns mean — a consistent offset is correctable, a variable one is not, which is what Table II addresses.]

D. Movement Quality Measures

Table III and Fig. 2 report agreement for the eleven movement-quality measures. [TODO: Prose: six measures reach $r_s \geq 0.86$. Total movement time is phase-derived and, because both systems share OMC phases here, is near tautological at $r_s = 1.00$ — report it as a consistency check, not as validation of pose. Number of movement units and interjoint coordination are intrinsically noisy; shoulder flexion is limited by the occluded seated-hip keypoint. Be explicit that these are floors of the construction, not of the reconstruction.]

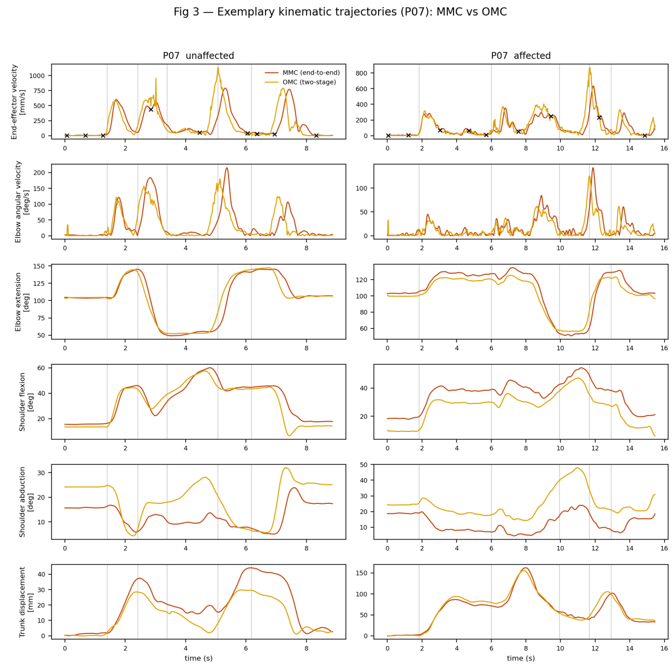

Fig. 1. Exemplary kinematic trajectories for the affected and unaffected arm, MMC versus OMC, with drinking-task phases marked and movement units indicated on the end-effector velocity. [TODO: name participant and trials]

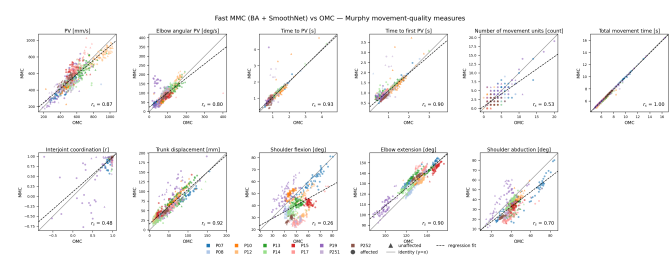

Fig. 2. Movement-quality measures, MMC versus OMC, per participant and arm. Eleven measures; see Sec. III-E for the two omitted by construction.

E. Ablations

[TODO: What each accelerated or simplified component costs in accuracy. Material already exists for: bundle adjustment with and without the bone-length prior; smoothing on and off; camera-dropout and miscalibration augmentation (augment*.csv); segmentation factors (seg_factors.csv). Select rather than dump — one table of the choices a reimplementer must get right.]

V. Discussion

[TODO: What the numbers support and what they do not. Our trajectory correlations sit a notch below IK-based markerless work (elbow extension $r \approx 0.96$ versus 0.99) — same ordering, similar ballpark, from consumer webcams at 60 Hz with no biomechanical model, and fast enough to report during the session. Where the residual error lives: the weakest measures are limited by occluded seated

TABLE I
Medians and [IQRs] for RMSE, r, bias, and time lag of kinematic trajectories, calculated across all participants and differentiating between the affected and unaffected arms.

Kinematic	Value	Unaffected arm	Affected arm
End-effector velocity	r	0.75 [0.65, 0.84]	0.77 [0.71, 0.82]
	RMSE (mm/s)	141.26 [123.01, 173.38]	133.58 [116.44, 156.92]
	Bias (mm/s)	-24.98 [-30.07, -15.12]	-23.90 [-31.77, -12.13]
	Time lag (s)	0.02 [0.00, 0.13]	0.03 [0.00, 0.15]
Elbow angular velocity	r	0.75 [0.65, 0.84]	0.74 [0.65, 0.81]
	RMSE (deg/s)	28.28 [23.31, 35.12]	26.68 [21.11, 32.93]
	Bias (deg/s)	2.02 [0.95, 3.03]	2.31 [1.28, 3.19]
	Time lag (s)	0.02 [0.00, 0.13]	0.03 [0.00, 0.17]
Elbow extension	r	0.96 [0.91, 0.97]	0.96 [0.94, 0.98]
	RMSE (deg)	9.06 [7.09, 11.84]	9.18 [7.22, 10.95]
	Bias (deg)	0.55 [-2.12, 4.20]	-0.31 [-5.13, 3.14]
	Time lag (s)	0.02 [0.00, 0.13]	0.03 [0.00, 0.17]
Shoulder flexion	r	0.87 [0.81, 0.93]	0.90 [0.77, 0.93]
	RMSE (deg)	7.79 [5.29, 13.37]	8.14 [5.40, 11.73]
	Bias (deg)	-2.87 [-8.82, 4.67]	-1.57 [-11.93, 3.01]
	Time lag (s)	0.02 [0.00, 0.13]	0.03 [0.00, 0.17]
Shoulder abduction	r	0.50 [0.17, 0.75]	0.70 [0.56, 0.84]
	RMSE (deg)	4.78 [3.75, 7.39]	5.19 [3.45, 6.14]
	Bias (deg)	-8.77 [-11.96, -6.40]	-7.53 [-10.57, -5.33]
	Time lag (s)	0.02 [0.00, 0.13]	0.03 [0.00, 0.17]
Trunk displacement	r	0.86 [0.61, 0.94]	0.93 [0.66, 0.96]
	RMSE (mm)	7.75 [5.73, 10.64]	7.32 [6.17, 10.29]
	Bias (mm)	8.93 [5.14, 14.37]	10.65 [7.21, 14.18]
	Time lag (s)	0.02 [0.00, 0.13]	0.03 [0.00, 0.17]

TABLE II
Mean IQR of the static bias across participants, i.e. the within-participant trial-to-trial variability of the offset between MMC and OMC.

Kinematic	Unaffected arm	Affected arm
End-effector velocity (mm/s)	7.31	10.88
Elbow angular velocity (deg/s)	2.05	1.48
Elbow extension (deg)	2.68	1.63
Shoulder flexion (deg)	3.31	2.66
Shoulder abduction (deg)	1.55	1.21
Trunk displacement (mm)	4.10	4.73

TABLE III
Movement-quality measures: Spearman correlation between MMC and OMC across all trials (r_s) and on per-participant, per-arm averaged trials (r_{av}).

Movement quality measure	r_s	r_{av}	n
PV [mm/s]	0.87	0.93	697
Elbow angular PV [deg/s]	0.80	0.77	695
Time to PV [s]	0.93	0.93	697
Time to first PV [s]	0.90	0.94	697
Number of movement units [n]	0.53	0.72	697
Total movement time [s]	1.00	1.00	697
Interjoint coordination	0.48	0.66	695
Trunk displacement [mm]	0.92	0.94	695
Shoulder flexion [deg]	0.26	0.19	695
Elbow extension [deg]	0.90	0.88	695
Shoulder abduction [deg]	0.70	0.68	695

anatomy, which is a camera-placement and modeling issue rather than a detector one. Clinical implication of same-session feedback.]

VI. Limitations

[TODO: Cohort size and composition; single site; the trajectory subset noted in Methods; dependence on camera placement and calibration quality, including that some sessions required re-cutting or recalibration before they were usable; measures that could not be validated for lack of a ground-truth column; and that geometric angles are not anatomical joint angles.]

VII. Conclusion

[TODO: Two or three sentences.]

Acknowledgment

[TODO: funding, DELTA study data providers]

References

[1] M. A. Murphy, C. Häger-Ross, M. A. Murphy, H. C. Persson, and K. S. Sunnerhagen, "Kinematic variables quantifying upper-extremity performance after stroke during reaching and drinking from a glass," Neurorehabilitation and Neural Repair, vol. 25, no. 1, pp. 71–80, 2011.

[2] T. Unger, A. Sal Moslehian, J. D. Peiffer, J. Ullrich, R. Gassert, O. Lambercy, R. J. Cotton, and C. Awai Easthope, "Differentiable biomechanics for markerless motion capture in upper limb stroke rehabilitation: A comparison with optical motion capture," arXiv preprint arXiv:2411.14992, 2024. [Online]. Available: https://arxiv.org/abs/2411.14992